

Practical 8: Intrusion Detection System

1. AIM: Set up and configure an intrusion detection system (IDS) to monitor network traffic and detect potential security breaches or malicious activities.

2. Software Requirements

- Ubuntu Linux 22.04/24.04
- Snort IDS
- Wireshark
- Nmap

3. System Requirements

- Minimum 4 GB RAM
- Minimum 20 GB free disk space
- Internet connection
- Administrator or sudo access
- One network interface connected to the network

4. Step 1: Update Ubuntu

```
C:\Windows\System32>wsl --install
Downloading: Windows Subsystem for Linux 2.7.12
Installing: Windows Subsystem for Linux 2.7.12
Windows Subsystem for Linux 2.7.12 has been installed.
Installing Windows optional component: VirtualMachinePlatform

Deployment Image Servicing and Management tool
Version: 10.0.26100.8972

Image Version: 10.0.26200.9168

Enabling feature(s)
[=====100.0%=====]
The operation completed successfully.
The requested operation is successful. Changes will not be effective until the system is rebooted.
The requested operation is successful. Changes will not be effective until the system is rebooted.

C:\Windows\System32>
```

```
sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble

sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ uname -a
Linux DESKTOP-N1QWMS3 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux

sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ sudo apt update
[sudo] password for sanjana:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
```

```
161 packages can be upgraded. Run 'apt list --upgradable' to see them.
sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  apparmor apport apport-core-dump-handler binutils binutils-common binutils-x86-64-linux-gnu
  bsdxtrautils bsdtails byobu ca-certificates cloud-init console-setup console-setup-linux coreutils
  curl distro-info-data dpkg eject fdisk gawk gcc-14-base glib2.0-packagekit-glib-1.0 gzip iproute2
  keyboard-configuration kmod krb5-locales libapparmor1 libavahi-client3 libavahi-common-data
  libavahi-common3 libbinutils libblkid1 libc-bin libc6 libc-bin libc6 libcap2 libcap2-bin libctf-nobfd0 libctf0
  libcurl3t64-gnutls libcurl4t64 libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm2
  libegl-mesa0 libexpat1 libfdisk1 libfreeType6 libgbm1 libgcc-s1 libgcrpt20 libgdk-pixbuf-2.0-0
  libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgl1-mesa-dri libglx-mesa0 libgnutls30t64 libgprofng0
  libgraphite2-3 libgssapi-krb5-2 libk5crypto3 libkmod2 libkrb5-3 libkrb5support0 liblcms2-2 libllvm20
```

Step 2: Install Snort

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```
done.
sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ ping -c 4 google.com
PING google.com (192.178.174.113) 56(84) bytes of data:
64 bytes from lcbomo-in-f113.1e100.net (192.178.174.113): icmp_seq=1 ttl=112 time=5.75 ms
64 bytes from lcbomo-in-f113.1e100.net (192.178.174.113): icmp_seq=2 ttl=112 time=5.24 ms
64 bytes from lcbomo-in-f113.1e100.net (192.178.174.113): icmp_seq=3 ttl=112 time=5.37 ms
64 bytes from lcbomo-in-f113.1e100.net (192.178.174.113): icmp_seq=4 ttl=112 time=8.01 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3105ms
rtt min/avg/max/mdev = 5.237/6.091/8.006/1.121 ms
sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ sudo apt install snort -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libverbs-providers libauthen-sasl-perl libclone-perl libdaq2t64 libdata-dump-perl libdumbnet1
  libencode-locale-perl libfile-listing-perl libfont-afm-perl libhtml-form-perl libhtml-format-perl
  libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl libhttp-cookies-perl libhttp-daemon-perl
  libhttp-date-perl libhttp-message-perl libhttp-negotiate-perl libibverbs1 libio-html-perl
```

Step 3: Check the Network Interface

```
Processing triggers for man-db (2.12.0-4build2) ...
sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ snort -V

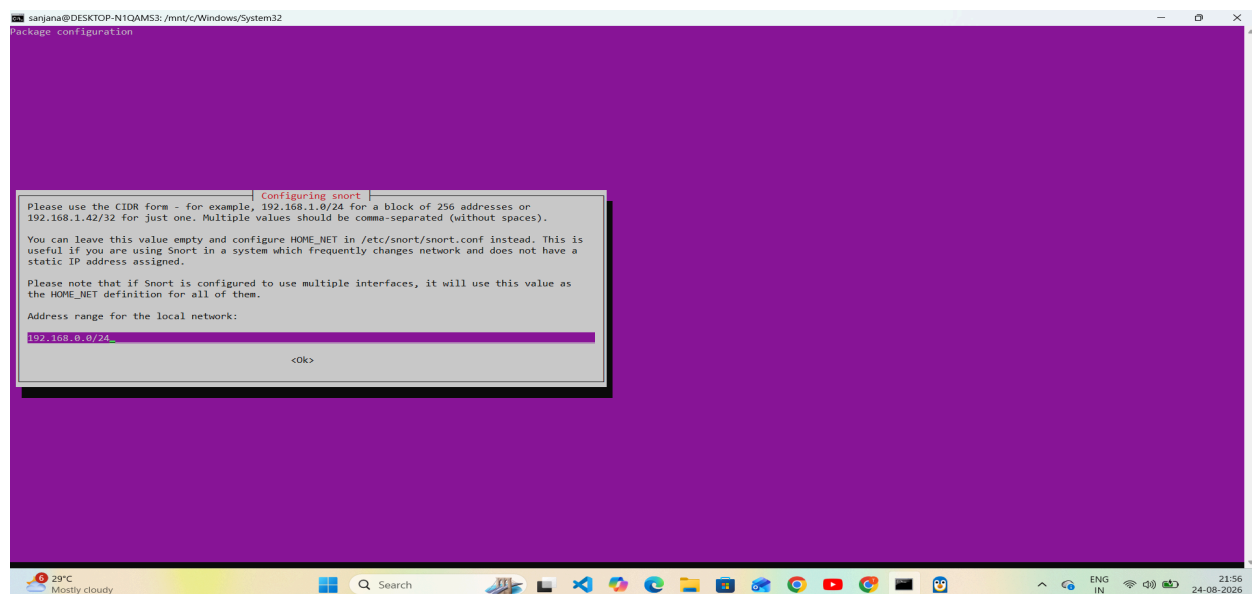
-> Snort! <-
o'-- Version 2.9.20 GRE (Build 82) x86_64
...- By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
    Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
    Copyright (C) 1998-2013 Sourcefire, Inc., et al.
    Using libpcap version 1.10.4 (with TPACKET_V3)
    Using PCRE version: 8.39 2016-06-14
    Using ZLIB version: 1.3

sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
```

Step 4: Test Snort Configuration

```
sanjana@DESKTOP-N1QWMS3:/mnt/c/Windows/System32$ sudo snort -T -c /etc/snort/snort.conf
[sudo] password for sanjana:
Running in Test mode

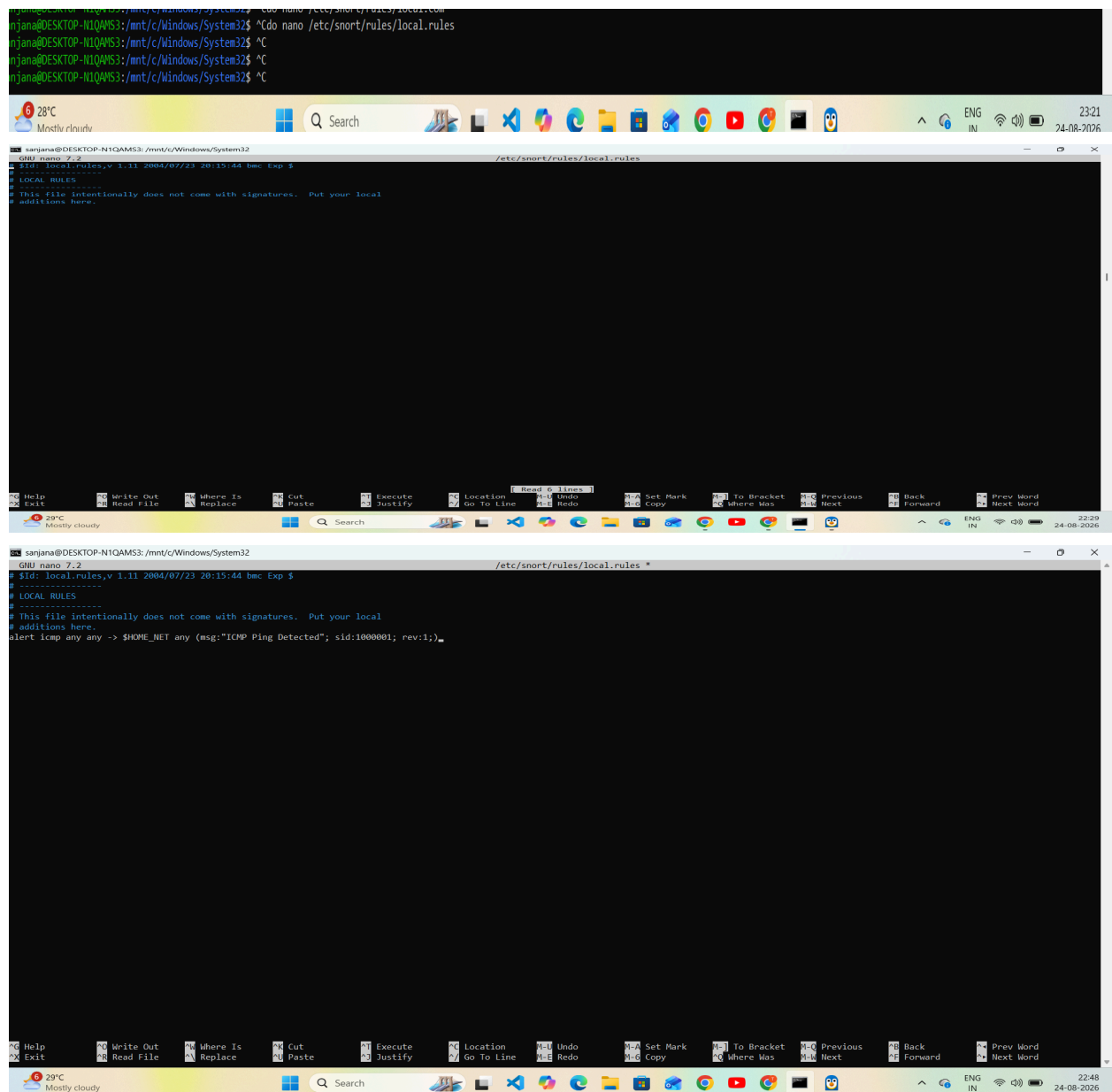
---= Initializing Snort ===
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : { 80:81 311 383 591 593 901 1220 1414 1741 1830 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8...
```



Step 5: Start Snort in Network Monitoring Mode

Step 6: Create a Custom IDS Rule

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The image consists of two screenshots of a Windows terminal window, showing the configuration of a Snort rule in the file `/etc/snort/rules/local.rules`.

Top Screenshot: The terminal shows the command `nano /etc/snort/rules/local.rules` being executed. The file is opened in nano editor, showing the default Snort rule structure. The terminal output is as follows:

```
sanjana@DESKTOP-N1QAMS3: /mnt/c/Windows/System32$ ^Cdo nano /etc/snort/rules/local.rules
sanjana@DESKTOP-N1QAMS3: /mnt/c/Windows/System32$ ^C
sanjana@DESKTOP-N1QAMS3: /mnt/c/Windows/System32$ ^C
sanjana@DESKTOP-N1QAMS3: /mnt/c/Windows/System32$ ^C
```

Bottom Screenshot: The terminal shows the same file being edited. The rule is configured to alert on any ICMP ping detected. The terminal output is as follows:

```
sanjana@DESKTOP-N1QAMS3: /mnt/c/Windows/System32$ nano /etc/snort/rules/local.rules
GNU nano 7.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1);
```

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```
sanjana@DESKTOP-N1QW53: /mnt/c/Windows/System32$ sudo nano /etc/snort/rules/local.rules
# nano /etc/snort/rules/local.rules
# Local Rules
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

File Name to Write: /etc/snort/rules/local.rules
[OK] Format: Mac Format
[OK] Backup File: Backup File
```

Step 7: Ensure the Local Rule File Is Included

```
sanjana@DESKTOP-N1QW53: /mnt/c/Windows/System32$ sudo cat /etc/snort/rules/local.rules
[sudo] password for sanjana:
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

sanjana@DESKTOP-N1QW53: /mnt/c/Windows/System32$
```

Step 8: Test the Configuration Again

Step 9: Start IDS Monitoring

```
Total snort Fixed Memory Cost - MaxRss:106580
Snort successfully validated the configuration!
Snort exiting
sanjana@DESKTOP-N1QW53: /mnt/c/Windows/System32$ sudo snort -c /etc/snort/snort.conf -i eth0
Running in IDS mode

---= Initializing Snort =---
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined: [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined: [ 0-79 81:65535 ]
PortVar 'ORACLE_PORTS' defined: [ 1024:65535 ]
PortVar 'SSH_PORTS' defined: [ 22 ]
PortVar 'FTP_PORTS' defined: [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined: [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined: [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined: [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
  Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsf_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
Finished loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssl_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_modbus_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_smtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dce2_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dns_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sip_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_reputation_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssh_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_lmtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_pop_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_sdf_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ftptelnet_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dnp3_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_appid_preproc.so... done
```

Step 10: Generate Test Traffic

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```
sanjana@DESKTOP-N1QAMS3:~$ ping -c 4 172.17.206.91
PING 172.17.206.91 (172.17.206.91) 56(84) bytes of data.
64 bytes from 172.17.206.91: icmp_seq=1 ttl=64 time=4.53 ms
64 bytes from 172.17.206.91: icmp_seq=2 ttl=64 time=0.062 ms
64 bytes from 172.17.206.91: icmp_seq=3 ttl=64 time=0.076 ms
64 bytes from 172.17.206.91: icmp_seq=4 ttl=64 time=0.075 ms

--- 172.17.206.91 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3224ms
rtt min/avg/max/mdev = 0.062/1.185/4.529/1.930 ms
sanjana@DESKTOP-N1QAMS3:~$ |
```

```
Using ZLIB version: 1.3

Rules Engine: SF_SHORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Preprocessor Object: SF_SMP Version 1.1 <Build 9>
Preprocessor Object: SF_XOBBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLLPP Version 1.1 <Build 4>

Commencing packet processing (pid=14686)
```

Step 11: Test Port Scanning Detection

```
sanjana@DESKTOP-N1QAMS3:~$ sudo pkill snort
[sudo] password for sanjana:
sanjana@DESKTOP-N1QAMS3:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [969 kB]
Fetched 1095 kB in 1s (1072 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
sanjana@DESKTOP-N1QAMS3:~$ sudo apt install nmap -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libblas3 liblinear4 liblua5.4-0 libssh2-1t64 nmap-common
Suggested packages:
  liblinear-tools liblinear-dev ncat ndiff zenmap
The following NEW packages will be installed:
  libblas3 liblinear4 liblua5.4-0 libssh2-1t64 nmap nmap-common
0 upgraded, 6 newly installed, 0 to remove and 3 not upgraded.
Need to get 6454 kB of archives.
After this operation, 28.0 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libblas3 amd64 3.12.0-3build1.1 [238 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 liblinear4 amd64 2.3.0+dfsg-5build1 [42.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libssh2-1t64 amd64 1.11.0-4.1ubuntu0.24.04.3 [121 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap-common all 7.94+git20230807.3be01efb1+dfsg-3build2 [4192 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap amd64 7.94+git20230807.3be01efb1+dfsg-3build2 [1694 kB]
Fetched 6454 kB in 1s (7016 kB/s)
Selecting previously unselected package libblas3:amd64.
(Reading database ... 42024 files and directories currently installed.)
Preparing to unpack .../0-libblas3_3.12.0-3build1.1_amd64.deb ...
Unpacking libblas3:amd64 (3.12.0-3build1.1) ...
Selecting previously unselected package liblinear4:amd64.
Preparing to unpack .../1-liblinear4_2.3.0+dfsg-5build1_amd64.deb ...
Unpacking liblinear4:amd64 (2.3.0+dfsg-5build1) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../2-liblua5.4-0_5.4.6-3build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-3build2) ...
Selecting previously unselected package libssh2-1t64:amd64.
Preparing to unpack .../3-libssh2-1t64_1.11.0-4.1ubuntu0.24.04.3_amd64.deb ...
Unpacking libssh2-1t64:amd64 (1.11.0-4.1ubuntu0.24.04.3) ...
Selecting previously unselected package nmap-common.
Preparing to unpack .../4-nmap-common_7.94+git20230807.3be01efb1+dfsg-3build2_all.deb ...
Unpacking nmap-common (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Selecting previously unselected package nmap.
Preparing to unpack .../5-nmap_7.94+git20230807.3be01efb1+dfsg-3build2_amd64.deb ...
Unpacking nmap (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Setting up libblas3:amd64 (3.12.0-3build1.1) ...
Setting up liblinear4:amd64 (2.3.0+dfsg-5build1) ...
Setting up liblua5.4-0:amd64 (5.4.6-3build2) ...
Setting up libssh2-1t64:amd64 (1.11.0-4.1ubuntu0.24.04.3) ...
Setting up nmap-common (7.94+git20230807.3be01efb1+dfsg-3build2) ...
Setting up nmap (7.94+git20230807.3be01efb1+dfsg-3build2) ...
```

```
sanjana@DESKTOP-N1QAMS3:~$ nmap --version
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.0.13 libssh2-1.11.0 libz-1.3 libpcap-1.10.4 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
sanjana@DESKTOP-N1QAMS3:~$ sudo nano /etc/snort/rules/local.rules
```

Step 12: View IDS Alerts

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```
sanjana@DESKTOP-N1QAMS3: ~  
GNU nano 7.2 /etc/snort/rules/local.rules  
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
#  
# LOCAL RULES  
#  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)  
alert tcp any any -> $HOME_NET any (flags:S, msg:"TCP SYN Scan Detected"; sid:1000002; rev:1;)  
  
File Name to Write: /etc/snort/rules/local.rules  
[M-D] DOS Format [M-A] Append [M-B] Backup File  
[M-M] Mac Format [M-P] Prepend [B] Browse  
[H] Help [C] Cancel  
  
sanjana@DESKTOP-N1QAMS3:~$ sudo cat /etc/snort/rules/local.rules  
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $  
#  
# LOCAL RULES  
#  
# This file intentionally does not come with signatures. Put your local  
# additions here.  
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)  
alert tcp any any -> $HOME_NET any (flags:S, msg:"TCP SYN Scan Detected"; sid:1000002; rev:1;)  
sanjana@DESKTOP-N1QAMS3:~$  
  
sanjana@DESKTOP-N1QAMS3:~$ sudo snort -T -c /etc/snort/snort.conf  
Running in Test mode  
  
--== Initializing Snort ==--  
Initializing Output Plugins!  
Initializing Preprocessors!  
Initializing Plug-ins!  
Parsing Rules file "/etc/snort/snort.conf"  
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7  
777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34444:34444 4108  
0 50002 55555 ]  
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]  
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]  
PortVar 'SSH_PORTS' defined : [ 22 ]  
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]  
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]  
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 714  
4:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 344  
43:34444 41080 50002 55555 ]  
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]  
Detection:  
  Search-Method = AC-Full-Q  
  Split Any/Any group = enabled  
  Search-Method-Optimizations = enabled  
  Maximum pattern length = 20  
Tagged Packet Limit: 256  
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsfe_engine.so... done  
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...  
  
Total snort Fixed Memory Cost - MaxRss:104988  
Snort successfully validated the configuration!  
Snort exiting
```

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```
sanjana@DESKTOP-N1QAMS3:~$ sudo snort -c /etc/snort/snort.conf -i eth0
Running in IDS mode

---= Initializing Snort ==---
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7
777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 4108
0 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 714
4:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 344
43:34444 41080 50002 55555 ]
PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsfe_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
```

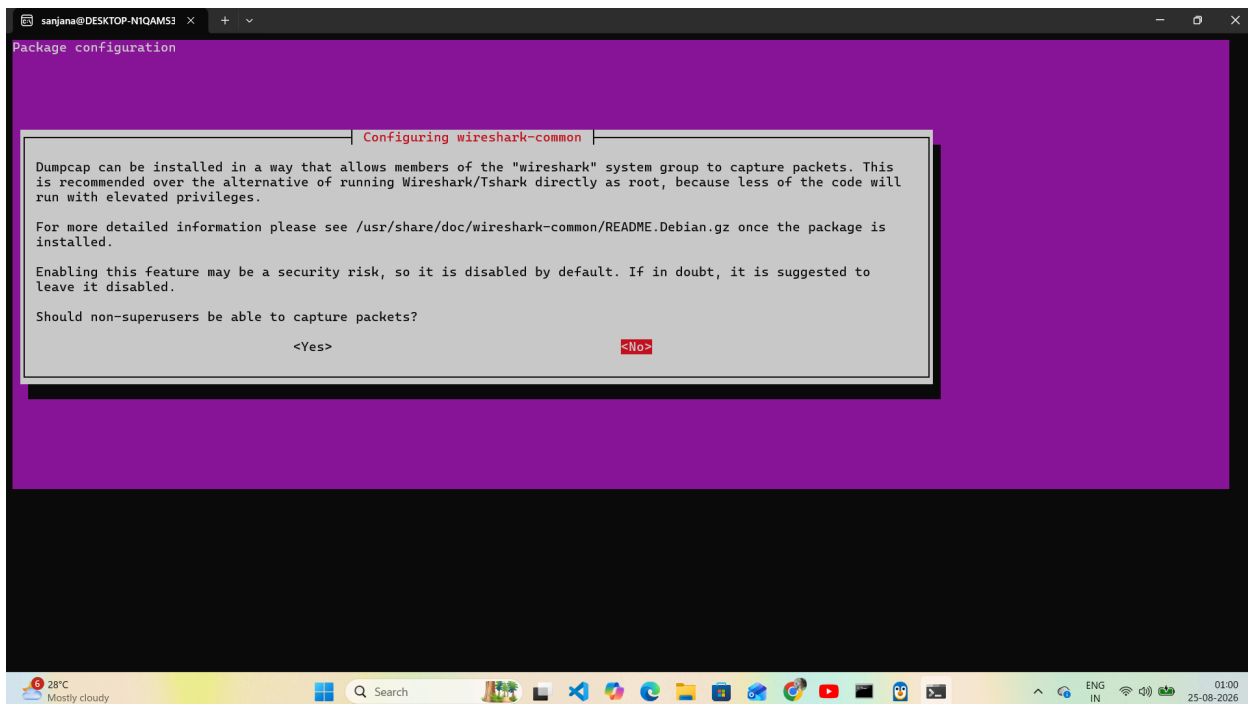
```
sanjana@DESKTOP-N1QAMS3:~$ sudo nmap -sS 172.17.206.91
[sudo] password for sanjana:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-24 19:22 UTC
Nmap scan report for 172.17.206.91
Host is up (0.0000030s latency).
All 1000 scanned ports on 172.17.206.91 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
sanjana@DESKTOP-N1QAMS3:~$
```

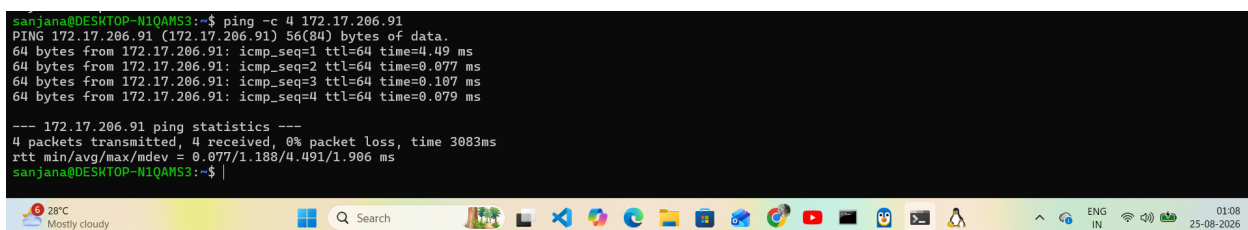
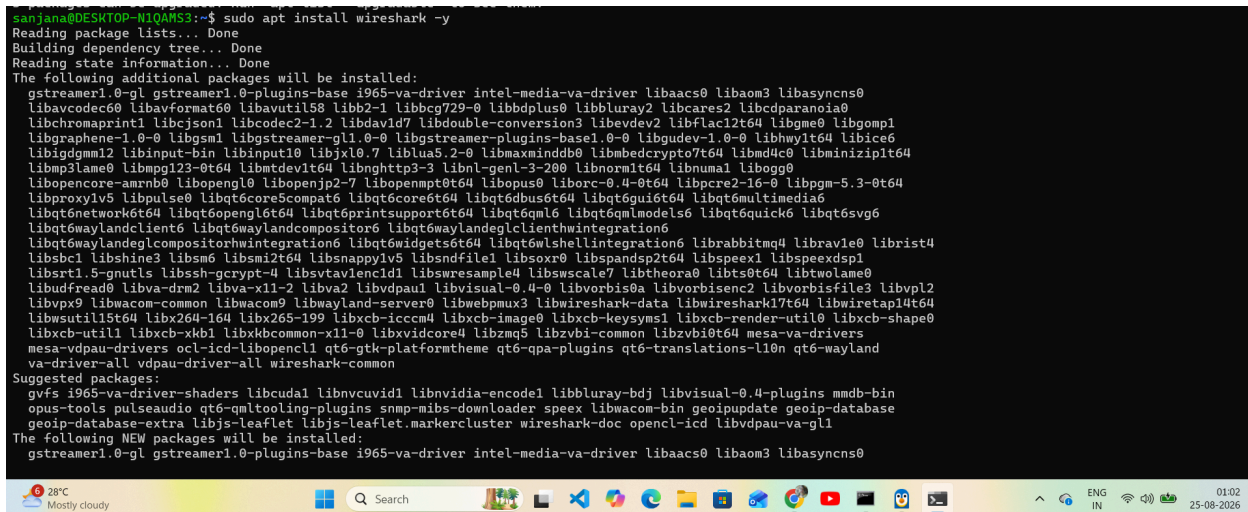
16. Step 13: Monitor Traffic Using Wireshark

```
sanjana@DESKTOP-N1QAMS3:~$ sudo pkill snort
[sudo] password for sanjana:
sanjana@DESKTOP-N1QAMS3:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Practical 8: Intrusion Detection System



Step 13: Monitor Traffic Using Wireshark



Practical 8: Intrusion Detection System

```
sanjana@DESKTOP-N1QAMS3:~$ sudo nmap -sS 172.17.206.91
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-24 19:40 UTC
Nmap scan report for 172.17.206.91
Host is up (0.0000030s latency).
All 1000 scanned ports on 172.17.206.91 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds
sanjana@DESKTOP-N1QAMS3:~$
```

```
Administrator: Command Prompt

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

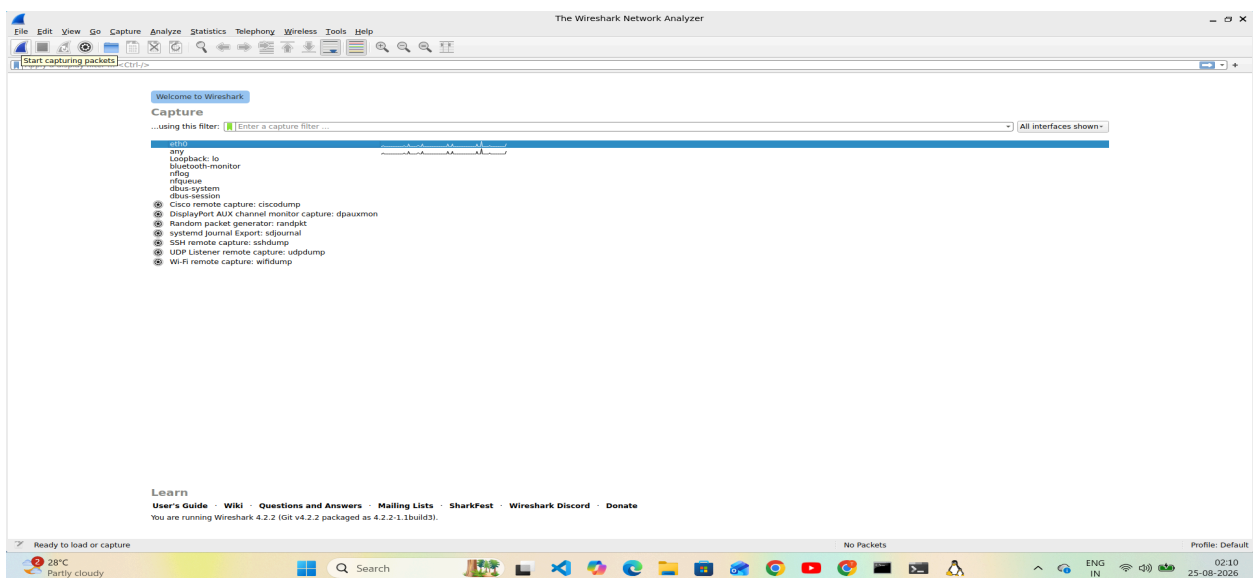
C:\Windows\System32>wsl --shutdown
```

```
sanjana@DESKTOP-N1QAMS3:~$ groups
sanjana adm cdrom sudo dip plugdev users wireshark
sanjana@DESKTOP-N1QAMS3:~$ wireshark
libEGL warning: failed to get driver name for fd -1

libEGL warning: MESA-LOADER: failed to retrieve device information

libEGL warning: failed to get driver name for fd -1

MESA: error: ZINK: failed to choose pdev
libEGL warning: egl: failed to create dri2 screen
** (wireshark:412) 20:40:38.503748 [Capture MESSAGE] -- Capture Start ...
** (wireshark:412) 20:40:38.602734 [Capture MESSAGE] -- Capture started
** (wireshark:412) 20:40:38.602823 [Capture MESSAGE] -- File: "/tmp/wireshark_eth07TC9T3.pcapng"
```



Practical 8: Intrusion Detection System

The image shows a terminal window and a Wireshark network traffic analysis interface. The terminal window, titled 'sanjana@DESKTOP-NIQAMS3', displays the following commands and output:

```
sanjana@DESKTOP-NIQAMS3:~$ curl https://mvl.u.com
sanjana@DESKTOP-NIQAMS3:~$ curl https://mvl.u.com
<!DOCTYPE html><html><head><script>window.onload=function(){window.location.href="/lander"}</script></head></html>sanjana@DESKTOP-NIQAMS3:~$
```

The Wireshark interface, titled 'Capturing on eth0', shows a list of network packets. The selected packet is a TCP packet (No. 80) with the following details:

- Time: 80.354138127983
- Source: 76.223.54.146
- Destination: 172.17.206.91
- Protocol: TCP
- Length: 66
- Info: 66 443 -> 50500 [ACK] Seq=6549 Ack=675 Win=62880 Len=0 TSval=2518756443 TSecr=2518756443

The packet details pane shows the following information:

- Epoch Arrival Time: 1787064776.687114642
- Time shift for this packet: 0.000000000 seconds
- Time delta from previous captured frame: 0.000000000
- Time delta from previous displayed frame: 0.000000000
- Time since reference or first frame: 0.000000000
- Frame Number: 1
- Frame Length: 70 bytes (560 bits)
- Capture Length: 70 bytes (560 bits)
- Frame is marked: False
- Frame is ignored: False
- Protocols in frame: eth:ethertype:ipv6:icmpv6
- Coloring Rule Name: ICMP
- Coloring Rule String: icmp || icmpv6
- Ethernet II, Src: Microsoft 75:ae:40 (00:15:5d:75:ae:40), Destination: IP multicast 02 (33:33:00:00:00:02)
- Address: IP multicast 02 (33:33:00:00:00:02)
- Source: Microsoft 75:ae:40 (00:15:5d:75:ae:40)
- Address: Microsoft 75:ae:40 (00:15:5d:75:ae:40)
- Type: IPv6 (0x86dd)
- Internet Protocol Version 6, Src: fe80::215:5dff:fe010::, Version: 6
- Next Header: ICMPv6 (58)
- Hop Limit: 255
- Source Address: fe80::215:5dff:fe010::
- Destination Address: ff02::2
- Type: Router Solicitation (133)
- Checksum: 0x6398 (correct)
- Checksum Status: Good
- Reserved: 00000000
- ICMPv6 Option (Source Link-layer address : 00:15:5d:75:ae:40)
- Type: Source Link-layer address (1)
- Length: 1 (8 bytes)
- Link-layer address: Microsoft 75:ae:40 (00:15:5d:75:ae:40)

The bottom status bar of Wireshark shows 'Packets: 134 / Displayed: 134 (100.0%)' and 'Profile: Default'.